

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)**

I B.Tech I Semester Regular Examinations, January-2024

**BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
(Common to CSE, CSE(AIML), CSE(AI) and CSE(DS))**

Time: 3 hours

Max. Marks: 70

**PART- A
(BASIC ELECTRICAL ENGINEERING)**

Note:

- Question 1 shall contain 5 compulsory short answer questions such that each question carry 1 Mark.
- In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Max. Marks: 35

Q. No.	Question	BTL	CO	Marks
1.	a) Define Power factor.	K1	CO1	1M
	b) Mention the advantages of sinusoidal alternating quantity.	K1	CO1	1M
	c) What is the principle of operation of single phase transformer?	K1	CO2	1M
	d) Write the balance equation of Wheatstone Bridge?	K1	CO2	1M
	e) Write the Safety Precautions to avoid shock .	K1	CO3	1M
UNIT-I				
2.	a) State and explain Kirchoff's current law and voltage law with an example.	K1	CO1	5M
	b) A voltage of 200 V is applied to a series circuit consisting of a 20 ohms resistance and 30 mH inductance are connected in series and the circuit is fed from 230 V, 50 Hz AC supply. Find (i) Inductive reactance and total impedance (ii) current in the circuit (iii) voltage across resistance and inductance (iv) real, reactive and apparent power (v) Power factor.	K3	CO1	5M
OR				
3.	a) Deduce an expression for the equivalent capacitance of three capacitors connected in i) Parallel ii) Series.	K1	CO1	5M
	b) State and Explain Superposition Theorem.	K3	CO1	5M
UNIT-II				
4.	a) Explain how AC. voltage generated is converted to DC. voltage in a generator.	K2	CO2	5M
	b) Explain the construction of 3-phase Alternator.	K1	CO2	5M
OR				
5.	a) Explain the working principle 3-phase Induction motor.	K1	CO2	5M
	b) Explain the working principle of DC generator.	K1	CO2	5M
UNIT-III				
6.	a) Draw the Layout of Nuclear Power Generation Plant and explain its operation.	K2	CO3	5M
	b) Differentiate between Conventional and non-conventional energy resources.	K3	CO3	5M
OR				
7.	a) Explain the calculation of electricity bill for domestic consumers.	K3	CO3	4M
	b) Explain the Working principle of Miniature circuit breaker with its merits and demerits.	K1	CO3	6M

PART-B
(BASIC ELECTRONICS ENGINEERING)

Note:

- i. Question 1 shall contain 5 compulsory short answer questions such that each question carry 1 Mark.
ii. In each of the questions from 8 to the last question, there shall be either/or type questions of 10 marks each.

Max. Marks: 35

Q. No.	Question	BTL	CO	Marks
1.	f) What is the depletion region?	K1	CO4	1M
	g) Define rectifier.	K1	CO5	1M
	h) What is the application of Amplifier?	K1	CO5	1M
	i) Write first ten numbers in Base-16 system.	K1	CO6	1M
	j) What is the application of Flip-Flop?	K1	CO6	1M
UNIT-IV				
8.	a) Explain characteristics of PN Junction Diode.	K2	CO4	5M
	b) With neat diagram explain about Small signal CE amplifier.	K2	CO4	5M
OR				
9.	Explain about CB configuration and its characteristics with neat diagram	K2	CO4	10M
UNIT-V				
10	Draw the block diagram of electronic Instrumentation system and explain.	K2	CO5	10M
OR				
11.	a) Discuss the working principal of Zener Voltage regulator.	K2	CO5	5M
	b) Draw the block diagram of Public address system and explain.	K2	CO5	5M
UNIT-VI				
12.	a) Deduce X from the following? (i) $(BA0.C)_{16} = (X)_8$ (ii) $(10101100)_2 = (X)_{16}$	K3	CO6	5M
	b) Discuss Full adder with truth table	K3	CO6	5M
OR				
13.	a) Briefly explain any three Properties of Boolean algebra.	K2	CO6	5M
	b) What is Register? Discuss in detail about.	K2	CO6	5M